先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 12 – Semi-Supervised Learning

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Unlabeled data help only through an assumption



Every gain must be measured against a matched supervised baseline.

Guiding question

Which relation between input structure and class labels makes the unlabeled pool informative?

- smoothness: nearby points share labels
- cluster assumption: boundaries cross low density
- manifold assumption: labels vary along a low-dimensional structure
- multi-view assumption: distinct views provide complementary evidence

Six syllabus methods across three periods

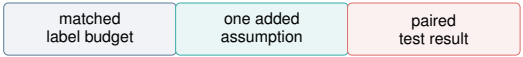
By the end

You can derive each objective, identify its structural assumption, implement representative algorithms, and design a leakage-safe comparison.

Period	Methods	Central question
1	self-training and multi-view	which pseudo-labels can be trusted?
2	generative models and S3VM	which distributional structure guides the boundary?
3	graph learning and clustering	which local constraints should propagate?

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Keep a supervised baseline visible



What to calculate

For every semi-supervised result, hold the true-label budget fixed and show whether unlabeled data improve or harm the same test cases.

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What unlabeled data can tell us

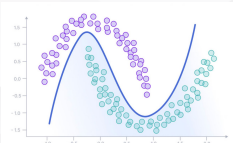
无标注数据有帮助?



只用标记数据的到的边界



根据所有数据聚类结果



半监督学习：同时使用标记数据和无标记数据

标记数据：正确性检查、类别数目、簇所属的类别
无标记数据：提供上下文信息、估计整个分布的形状

The assumptions behind semi-supervised learning

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半监督学习

基本思想：同时利用有标注数据和无标注数据学习

半监督分类/回归

- **给定:** 标注数据 $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2) \dots (\boldsymbol{x}_L, y_L)\}$，无标注数据 $\mathcal{D}_U = \{\boldsymbol{x}_{L+1}, \boldsymbol{x}_{L+2}, \dots \boldsymbol{x}_{L+U}\}$（$U \gg L$）
- 目标: 学习一个分类器 f 比只用标注数据学得更好

■ 半监督聚类/降维

- **给定:** 标注数据 $\{\boldsymbol{x}_i\}_{i=1}^N$, 但另外给出对数据的一些限制
  - 聚类: 两个点必须在一个簇, 或两个点一定不能在一个簇
  - 降维: 两个点降维后必须接近

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Semi-supervised learning setup: model and variables

Model and assumptions

Semi-supervised learning combines labeled pairs with unlabeled inputs under an explicit assumption about the data distribution.

$$\mathcal{D}_L = \{(x_i, y_i)\}_{i=1}^{\ell}, \qquad \mathcal{D}_U = \{x_j\}_{j=\ell+1}^{\ell+u}, \quad u \gg \ell$$

Symbol	Meaning	Status
$\mathcal{D}_L$	labeled sample	defines class semantics
$\mathcal{D}_U$	unlabeled sample	defines input structure
assumption	link between structure and labels	must be tested indirectly

Semi-supervised learning setup: worked example

Quantity	Value	Meaning
labeled rows	200	expert reviewed
unlabeled rows	20,000	same acquisition process
first baseline	supervised only	reference for every gain

What the result means

Use unlabeled data only after a supervised baseline and a domain-match check are in place.

Semi-supervised learning setup: what to check

Main idea

Unlabeled data constrain the shape of the input distribution; they do not reveal class meaning by themselves.

What to check

held-out labeled performance and calibration

Limit

a domain-shifted unlabeled pool can pull the decision boundary toward the wrong population

Semi-supervised learning setup: implementation

1. define label semantics
2. check train-unlabeled domain match
3. fit a supervised baseline
4. add one unlabeled-data mechanism
5. compare on untouched labels

Check the result

Check held-out labeled performance and calibration on training and held-out examples.

Common mistake

a domain-shifted unlabeled pool can pull the decision boundary toward the wrong population

Semi-supervised learning setup: common mistakes

Example

triage with 200 reviewed and 20,000 unreviewed studies

Explain the calculation and check held-out labeled performance and calibration, using a small example.

Common mistake

a domain-shifted unlabeled pool can pull the decision boundary toward the wrong population

What to do

Use unlabeled data only after a supervised baseline and a domain-match check are in place.

Semi-supervised learning setup: check your understanding

Question

When can unlabeled inputs improve a predictor trained from scarce labels?

Answer

Unlabeled data constrain the shape of the input distribution; they do not reveal class meaning by themselves.

Decision. Use unlabeled data only after a supervised baseline and a domain-match check are in place.

Self-training starts with the labeled examples

- Fit a classifier on the small labeled set L.
- Predict probabilities for examples in the unlabeled set U.
- Add selected predictions as pseudo-labels; refit the classifier.
- Pseudo-labels are model guesses and can reinforce mistakes.

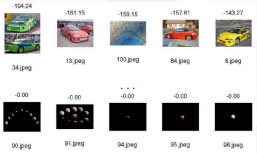
Self-training adds predicted labels

自动驾驶的例子: 图像分类

- 将最置信的图像及其预测标签加入到标注数据



- 重新训练分类器，重复上述过程



Self-training loop: model and variables

Model and assumptions

Self-training alternates between fitting a classifier and adding selected predictions from the unlabeled pool as pseudo-labels.

$$\widehat{y}_j = \arg\max_c p_\theta(c \mid x_j), \qquad \max_c p_\theta(c \mid x_j) \geq \tau$$

Symbol	Meaning	Status
$\widehat{y}_j$	pseudo-label	model generated
τ	acceptance threshold	controls quantity and precision
round	refit cycle	changes the training distribution

Self-training loop: worked example

Quantity	Value	Meaning
pool	10,000	unlabeled
above $\tau = 0.95$	620	candidate pseudo-labels
audited precision	97%	estimated from a small review

What the result means

Add only an audited, class-balanced subset and stop when held-out performance plateaus.

The choice rule for Self-training loop

Main idea

The learner changes its own future training set, so selection errors can compound across rounds.

What to check

pseudo-label precision by class and round

Limit

early false pseudo-labels can be reinforced until the model becomes confidently wrong

Self-training loop: implementation

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```
from sklearn.semi_supervised import SelfTrainingClassifier
from sklearn.linear_model import LogisticRegression
self_model = SelfTrainingClassifier(LogisticRegression(max_iter=1000), threshold=0.90, max_iter=8)
self_model.fit(X, y_ssl)
print('Self-training labeled count:', np.sum(self_model.transduction_ != -1))
```


Evidence and failure check

Check pseudo-label precision by class and round with the fitted model and data split; reject the run if early false pseudo-labels can be reinforced until the model becomes confidently wrong.

Self-training loop: common mistakes

Example

expanding labels for rare equipment faults

Explain the calculation and check pseudo-label precision by class and round, using a small example.

Common mistake

early false pseudo-labels can be reinforced until the model becomes confidently wrong

What to do

Add only an audited, class-balanced subset and stop when held-out performance plateaus.

Self-training loop: check your understanding

Question

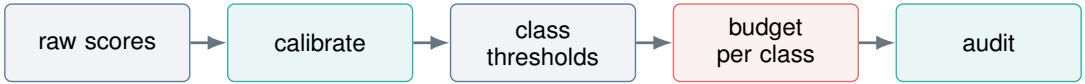
Which model predictions are reliable enough to become temporary labels?

Answer

The learner changes its own future training set, so selection errors can compound across rounds.

Decision. Add only an audited, class-balanced subset and stop when held-out performance plateaus.

Does a probability of 0.95 mean the same thing for every class?



Confidence ranks predictions only when the score is calibrated for the current domain and class prevalence.

Question

Does a probability of 0.95 mean the same thing for every class?

Example

pseudo-labeling a highly imbalanced inspection stream

Main idea

Confidence ranks predictions only when the score is calibrated for the current domain and class prevalence.

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A confident prediction can still be wrong

- Example: predicted probabilities $(0.97, 0.03)$ versus $(0.55, 0.45)$.
- A threshold of 0.95 accepts the first prediction and rejects the second.
- High confidence does not prove correctness: check calibration on labeled validation data.
- Monitor accepted pseudo-labels by class, including minority classes.



Confidence, calibration, and balance: model and variables

Model and assumptions

A pseudo-label policy combines calibrated confidence, class balance, uncertainty, and round-specific budgets.

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{n} \left|\mathrm{acc}(I_b) - \mathrm{conf}(I_b)\right|$$

Symbol	Meaning	Status
I_b	confidence bin	validation predictions
accuracy	empirical correctness	requires labels
confidence	mean predicted probability	model output

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Confidence, calibration, and balance: worked example

Quantity	Value	Meaning
class A threshold	0.90	96% precision
class B threshold	0.98	95% precision
single threshold	0.90	too many B errors

What the result means

Use class-specific thresholds when calibration or prevalence differs across classes.

Confidence, calibration, and balance: checking the result

Main idea

Confidence ranks predictions only when the score is calibrated for the current domain and class prevalence.

What to check

expected calibration error and classwise pseudo-label precision

Limit

a globally high confidence score can hide severe overconfidence for a minority class

What to record when running Confidence, calibration, and balance

1. reserve a calibration split
2. plot reliability by class
3. set thresholds for target precision
4. limit additions per round
5. recalibrate after domain change

Check the result

Check expected calibration error and classwise pseudo-label precision on training and held-out examples.

Common mistake

a globally high confidence score can hide severe overconfidence for a minority class

The warning sign for Confidence, calibration, and balance

Example

pseudo-labeling a highly imbalanced inspection stream

Explain the calculation and check expected calibration error and classwise pseudo-label precision, using a small example.

Common mistake

a globally high confidence score can hide severe overconfidence for a minority class

What to do

Use class-specific thresholds when calibration or prevalence differs across classes.

Explain Confidence, calibration, and balance without naming the algorithm

Question

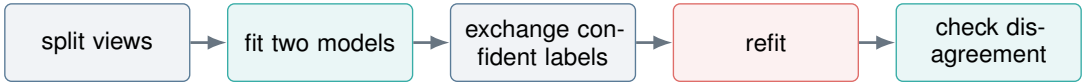
Does a probability of 0.95 mean the same thing for every class?

Answer

Confidence ranks predictions only when the score is calibrated for the current domain and class prevalence.

Decision. Use class-specific thresholds when calibration or prevalence differs across classes.

Can two feature views teach each other without sharing the same mistakes?



The benefit depends on each view being sufficient enough to predict and diverse enough to make different errors.

Question

Can two feature views teach each other without sharing the same mistakes?

Example

web-page classification from text and link structure

Main idea

The benefit depends on each view being sufficient enough to predict and diverse enough to make different errors.

Co-training uses two views of the same example

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协同训练（Co-training）

- 协同训练是自训练到多个监督分类器的扩展
- 协同训练的基本思想：
  - 在标注数据上迭代训练多个分类器
  - 多个分类器互相教对方：将各自最有信心的预测加到其他分类器的标注数据
- 协同训练中基础学习器的预测必须相关性不太强：多样性标准
- 为了促进分类器的多样性
  - 数据的多个不同视图：多视图学习
  - 学习算法本身的多样性：不同超参数的分类器、不同的基础分类器

How Co-training with multiple views changes the representation

Model and assumptions

Co-training fits one learner per view and lets confident predictions from one view enlarge the labeled set for the other.

$$p(y \mid x^{(1)}, x^{(2)}) \propto p(y)\, p(x^{(1)} \mid y)\, p(x^{(2)} \mid y)$$

Symbol	Meaning	Status
$x^{(1)}$	first view	one feature family
$x^{(2)}$	second view	another feature family
agreement	cross-view evidence	not guaranteed independence

Check the direction of change in Co-training with multiple views

Quantity	Value	Meaning
view 1	image texture	88% accuracy
view 2	metadata	82% accuracy
error correlation	0.18	useful diversity

What the result means

Use co-training only when each view has signal and their errors are demonstrably different.

When Co-training with multiple views earns a place in the pipeline

Main idea

The benefit depends on each view being sufficient enough to predict and diverse enough to make different errors.

What to check

cross-view error correlation and held-out gain

Limit

two highly correlated views can exchange the same wrong pseudo-labels and create false confirmation

Co-training with multiple views: implementation

1. define semantically distinct views
2. measure standalone sufficiency
3. estimate error correlation
4. exchange only audited labels
5. stop on validation evidence

Check the result

Check cross-view error correlation and held-out gain on training and held-out examples.

Common mistake

two highly correlated views can exchange the same wrong pseudo-labels and create false confirmation

When the objective of Co-training with multiple views is not enough

Example

web-page classification from text and link structure

Explain the calculation and check cross-view error correlation and held-out gain, using a small example.

Common mistake

two highly correlated views can exchange the same wrong pseudo-labels and create false confirmation

What to do

Use co-training only when each view has signal and their errors are demonstrably different.

A final decision on Co-training with multiple views

Question

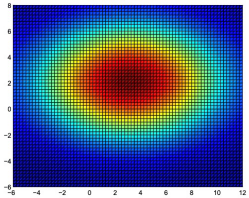
Can two feature views teach each other without sharing the same mistakes?

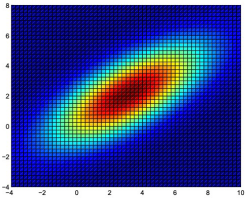
Answer

The benefit depends on each view being sufficient enough to predict and diverse enough to make different errors.

Decision. Use co-training only when each view has signal and their errors are demonstrably different.

Can the marginal density of all inputs help estimate class-conditional structure?





Question

Can the marginal density of all inputs help estimate class-conditional structure?

Example

two partially overlapping Gaussian populations

Main idea

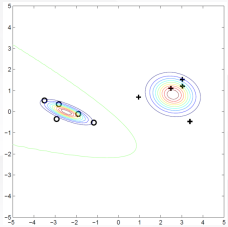
Unlabeled observations refine component shape and prevalence, while labeled observations attach class meaning to components.

Unlabeled data change the fitted density model

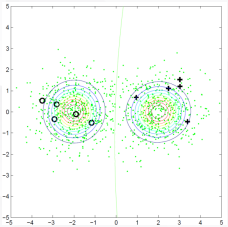
例：生成式模型

决策边界的不同，是由于模型最大化的目标不同

$p(\boldsymbol{X}_L, \boldsymbol{y}_L|\theta)$



$p(\boldsymbol{X}_L, \boldsymbol{y}_L, \boldsymbol{X}_U|\theta)$



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The variables that make Generative mixture model work

Model and assumptions

A generative semi-supervised model fits a joint distribution for inputs and labels using labeled and unlabeled likelihood terms.

$$\ell(\theta) = \sum_{i \in L} \log p_\theta(x_i, y_i) + \sum_{j \in U} \log \sum_{y} p_\theta(x_j, y)$$

Symbol	Meaning	Status
L	labeled indices	joint likelihood
U	unlabeled indices	marginal likelihood
θ	mixture parameters	shared across both terms

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Read the numbers produced by Generative mixture model

Quantity	Value	Meaning
components	2	shared covariance
labeled points	20	anchor component names
unlabeled points	980	refine means and covariance

What the result means

Use the unlabeled likelihood only when density fit and class alignment both pass diagnostics.

Three conditions for trusting Generative mixture model

Main idea

Unlabeled observations refine component shape and prevalence, while labeled observations attach class meaning to components.

What to check

held-out log loss and component-label alignment

Limit

a misspecified density can fit the input distribution well while moving the class boundary in the wrong direction

An implementation checklist for Generative mixture model

```
from sklearn.mixture import GaussianMixture
gmm = GaussianMixture(n_components=2, covariance_type='full', random_state=7)
gmm.fit(X)
resp = gmm.predict_proba(X)
print('GMM mean max responsibility:', resp.max(axis=1).mean().round(3))
```


Evidence and failure check

Check held-out log loss and component-label alignment with the fitted model and data split; reject the run if a misspecified density can fit the input distribution well while moving the class boundary in the wrong direction.

Reject Generative mixture model under this condition

Example

two partially overlapping Gaussian populations

Explain the calculation and check held-out log loss and component-label alignment, using a small example.

Common mistake

a misspecified density can fit the input distribution well while moving the class boundary in the wrong direction

What to do

Use the unlabeled likelihood only when density fit and class alignment both pass diagnostics.

Generative mixture model: check your understanding

Question

Can the marginal density of all inputs help estimate class-conditional structure?

Answer

Unlabeled observations refine component shape and prevalence, while labeled observations attach class meaning to components.

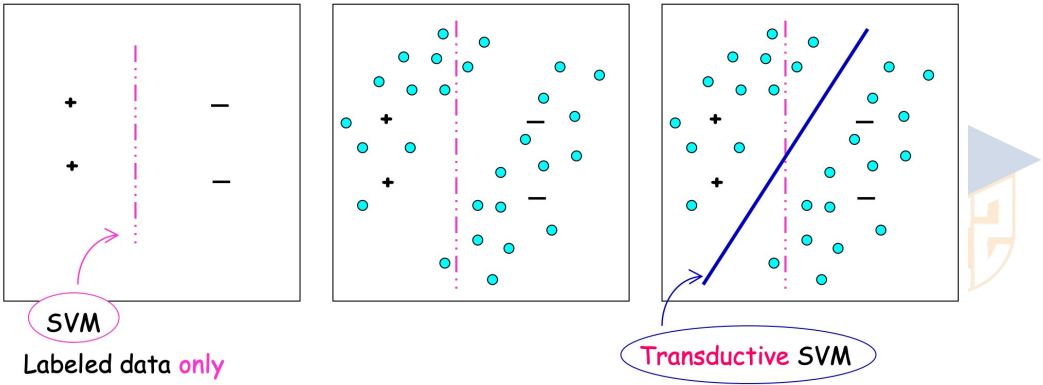
Decision. Use the unlabeled likelihood only when density fit and class alignment both pass diagnostics.

Unlabeled points influence where a classifier puts its boundary

- Standard SVM uses labeled points to fit a large-margin separator.
- Semi-supervised SVM also encourages unlabeled points to lie away from the boundary.
- This helps when high-density regions correspond to classes.
- The unlabeled-label assignment makes the problem generally nonconvex.



Place the boundary in a low-density region



Unlabeled inputs can move a separator toward a low-density region when the cluster assumption fits.

Semi-supervised SVM: model and variables

Model and assumptions

A transductive or semi-supervised SVM searches for labels on unlabeled points and a large-margin separator jointly.

$$\min_{w,b,\widehat{y}_U} \frac{1}{2}\|w\|^2 + C_L \sum_{i \in L} \xi_i + C_U \sum_{j \in U} \xi_j$$

Symbol	Meaning	Status
$\widehat{y}_U$	unknown transductive labels	discrete optimization
C_L	labeled penalty	trust in known labels
C_U	unlabeled penalty	strength of cluster assumption

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Semi-supervised SVM: worked example

Quantity	Value	Meaning
$C_U = 0$	supervised SVM	ignores pool
small C_U	gentle steering	safer continuation
large C_U	strong cluster prior	risk under overlap

What the result means

Increase unlabeled influence gradually and reject it if labeled validation loss rises.

A decision card for Semi-supervised SVM

Main idea

The low-density separation assumption is useful only when class boundaries avoid dense regions of the input distribution.

What to check

validation margin error and density along the boundary

Limit

when classes overlap in a dense region, low-density separation can force a confidently wrong boundary

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Reproduce the result of Semi-supervised SVM

1. fit a supervised SVM
2. inspect density near its margin
3. increase unlabeled weight gradually
4. track label switches
5. select with labeled validation data

Check the result

Check validation margin error and density along the boundary on training and held-out examples.

Common mistake

when classes overlap in a dense region, low-density separation can force a confidently wrong boundary

A stress test for Semi-supervised SVM

Example

two curved classes with few labels

Explain the calculation and check validation margin error and density along the boundary, using a small example.

Common mistake

when classes overlap in a dense region, low-density separation can force a confidently wrong boundary

What to do

Increase unlabeled influence gradually and reject it if labeled validation loss rises.

Semi-supervised SVM: check your understanding

Question

Can unlabeled points steer a maximum-margin boundary toward low-density regions?

Answer

The low-density separation assumption is useful only when class boundaries avoid dense regions of the input distribution.

Decision. Increase unlabeled influence gradually and reject it if labeled validation loss rises.

Which edges are trustworthy enough to carry label information?



Graph construction is the model: scaling, distance, neighbor count, and edge weights determine which labels can interact.

Question

Which edges are trustworthy enough to carry label information?

Example

propagating labels over patient similarity

Main idea

Graph construction is the model: scaling, distance, neighbor count, and edge weights determine which labels can interact.

Build a graph from nearby observations

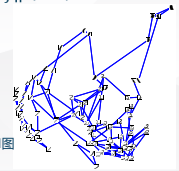
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构造相似图

- 节点: $\boldsymbol{X}_L \cup \boldsymbol{X}_U$
- 边：权重是基于特征来计算相邻节点之间的相似度，例如,
  - K最近邻图, 权重
  - 全连接图, 权重随距离衰减：$w_{i,j} = \exp(-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2/\sigma^2)$

两个点距离$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|$越远，$w_{i,j}$呈指数级下降

  - ε-半径(ε-radius)图



对称的2NN图

Neighborhood graph construction: model and variables

Model and assumptions

Graph methods represent every observation as a node and encode local similarity with weighted edges.

$$W_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_k(i)]$$

Symbol	Meaning	Status
W_{ij}	edge weight	local similarity
σ	kernel scale	controls weight decay
$\mathcal{N}_k(i)$	neighbor set	controls connectivity

Neighborhood graph construction: worked example

Quantity	Value	Meaning
$k = 2$	four components	labels cannot cross
$k = 10$	one stable graph	useful propagation
$k = 80$	many shortcuts	oversmoothing risk

What the result means

Treat neighborhood size as a structural hyperparameter and reject graphs with unstable connectivity.

Neighborhood graph construction: checking the result

Main idea

Graph construction is the model: scaling, distance, neighbor count, and edge weights determine which labels can interact.

What to check

graph connectivity, degree distribution, and labeled-edge purity

Limit

shortcuts between classes cause labels to leak across the true boundary

Neighborhood graph construction: implementation

1. fit scaling on training data
2. build several candidate graphs
3. check components and degree distribution
4. inspect cross-class edges on labeled nodes
5. choose a stable neighborhood range

Check the result

Check graph connectivity, degree distribution, and labeled-edge purity on training and held-out examples.

Common mistake

shortcuts between classes cause labels to leak across the true boundary

What breaks first in Neighborhood graph construction

Example

propagating labels over patient similarity

Explain the calculation and check graph connectivity, degree distribution, and labeled-edge purity, using a small example.

Common mistake

shortcuts between classes cause labels to leak across the true boundary

What to do

Treat neighborhood size as a structural hyperparameter and reject graphs with unstable connectivity.

Neighborhood graph construction: check your understanding

Question

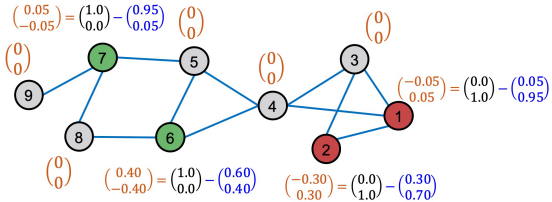
Which edges are trustworthy enough to carry label information?

Answer

Graph construction is the model: scaling, distance, neighbor count, and edge weights determine which labels can interact.

Decision. Treat neighborhood size as a structural hyperparameter and reject graphs with unstable connectivity.

How does a graph turn a few fixed labels into smooth predictions?



Question

How does a graph turn a few fixed labels into smooth predictions?

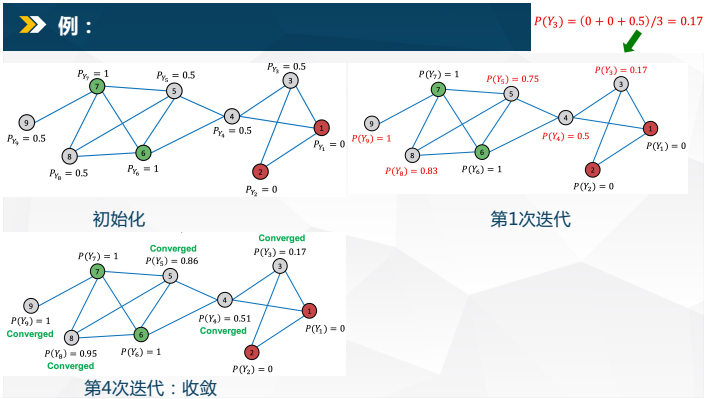
Example

classifying documents on a citation graph

Main idea

A harmonic solution assigns each unlabeled node a weighted average of its neighbors' current labels.

Labels spread along graph connections



Label propagation and harmonic solutions: model and variables

Model and assumptions

Label propagation minimizes variation across strong edges while clamping or softly anchoring known labels.

$$\min_F \frac{1}{2} \sum_{i,j} W_{ij} \|F_i - F_j\|^2, \qquad F_L = Y_L$$

Symbol	Meaning	Status
F_i	soft label vector	estimated at each node
Y_L	known labels	fixed boundary condition
graph energy	edge disagreement	smoothness objective

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Label propagation and harmonic solutions: worked example

Quantity	Value	Meaning
node A	$[1, 0]$	fixed label
node B	$[0, 1]$	fixed label
middle node	$[0.6, 0.4]$	weighted harmonic score

What the result means

Report unanchored components separately rather than presenting their propagated labels as evidence.

Accept or reject Label propagation and harmonic solutions

Main idea

A harmonic solution assigns each unlabeled node a weighted average of its neighbors' current labels.

What to check

validation accuracy by graph component and propagation stability

Limit

an isolated unlabeled component has no label anchor and receives an arbitrary or prior-driven prediction

Label propagation and harmonic solutions: implementation

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```
from sklearn.semi_supervised import LabelPropagation
graph_model = LabelPropagation(kernel='knn', n_neighbors=9, max_iter=500)
graph_model.fit(X, y_ssl)
print('Label propagation score:', round(graph_model.score(X, y), 3))
```


Evidence and failure check

Check validation accuracy by graph component and propagation stability with the fitted model and data split; reject the run if an isolated unlabeled component has no label anchor and receives an arbitrary or prior-driven prediction.

The hidden assumption in Label propagation and harmonic solutions

Example

classifying documents on a citation graph

Explain the calculation and check validation accuracy by graph component and propagation stability, using a small example.

Common mistake

an isolated unlabeled component has no label anchor and receives an arbitrary or prior-driven prediction

What to do

Report unanchored components separately rather than presenting their propagated labels as evidence.

Label propagation and harmonic solutions: check your understanding

Question

How does a graph turn a few fixed labels into smooth predictions?

Answer

A harmonic solution assigns each unlabeled node a weighted average of its neighbors' current labels.

Decision. Report unanchored components separately rather than presenting their propagated labels as evidence.

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Pairwise constraints add partial information to clustering

- Must-link: two examples should belong to the same cluster.
- Cannot-link: two examples should belong to different clusters.
- Example: two records of one person versus records known to be different people.
- Check constraint consistency and choose how to handle noisy constraints.

Check whether pairwise constraints can be satisfied

- Must-link (A, B) and must-link (B, C) imply that A, B, C share a cluster.
- Adding cannot-link (A, C) makes these hard constraints inconsistent.
- Check such conflicts before clustering.
- With uncertain constraints, use a stated soft penalty rather than pretending all are exact.



Semi-supervised clustering: model and variables

Model and assumptions

Constrained clustering combines an unsupervised partition objective with pairwise or partial-label constraints.

$$J = \sum_i \|x_i - \mu_{c_i}\|^2 + \lambda_M \sum_{(i,j)\in M} \mathbf{1}[c_i \neq c_j] + \lambda_C \sum_{(i,j)\in C} \mathbf{1}[c_i = c_j]$$

Symbol	Meaning	Status
M	must-link pairs	should share a cluster
C	cannot-link pairs	should differ
λ_M, λ_C	penalties	constraint strength

Semi-supervised clustering: worked example

Quantity	Value	Meaning
must-link	A with B	same cluster
cannot-link	B from C	different clusters
inconsistency	A with C	constraint conflict

What the result means

Audit constraints before fitting and report both satisfaction and partition stability.

Separate a good fit from a good decision in Semi-supervised clustering

Main idea

Constraints express local supervision; they do not guarantee a globally consistent or uniquely correct partition.

What to check

constraint satisfaction and unsupervised stability

Limit

noisy or contradictory pairwise constraints can make the optimization infeasible or distort every cluster

Semi-supervised clustering: implementation

1. verify constraint consistency
2. choose a base clustering objective
3. fit with constraint penalties
4. report violations
5. profile clusters using held-out attributes

Check the result

Check constraint satisfaction and unsupervised stability on training and held-out examples.

Common mistake

noisy or contradictory pairwise constraints can make the optimization infeasible or distort every cluster

Do not trust Semi-supervised clustering before this check

Example

grouping products with a few expert pairwise judgments

Explain the calculation and check constraint satisfaction and unsupervised stability, using a small example.

Common mistake

noisy or contradictory pairwise constraints can make the optimization infeasible or distort every cluster

What to do

Audit constraints before fitting and report both satisfaction and partition stability.

Semi-supervised clustering: check your understanding

Question

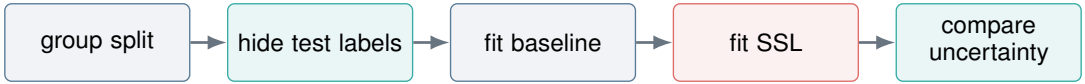
How should must-link and cannot-link information reshape a partition?

Answer

Constraints express local supervision; they do not guarantee a globally consistent or uniquely correct partition.

Decision. Audit constraints before fitting and report both satisfaction and partition stability.

Which labels may be used for tuning, and which must remain untouched?



The unlabeled pool may contain inputs from validation entities, but their labels and grouped identities must not influence training decisions.

Question

Which labels may be used for tuning, and which must remain untouched?

Example

comparing methods under a fixed annotation budget

Main idea

The unlabeled pool may contain inputs from validation entities, but their labels and grouped identities must not influence training decisions.

Compare methods using the same labeled-data budget

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- Use identical labeled training examples for the supervised and semi-supervised baselines.
- Keep validation and test labels out of pseudo-label construction.
- Repeat labeled-set sampling and report variability.
- Distinguish prediction on a fixed unlabeled pool from prediction on new examples.



Evaluation without label leakage: model and variables

Model and assumptions

A valid semi-supervised evaluation separates label acquisition, pseudo-label generation, hyperparameter tuning, and final testing.

$$\Delta = \text{score}(\text{SSL}) - \text{score}(\text{supervised baseline})$$

Symbol	Meaning	Status
Δ	incremental gain	same labeled test set
budget	number of true labels	held fixed across methods
entity group	leakage unit	split before pseudo-labeling

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Evaluation without label leakage: worked example

Quantity	Value	Meaning
label budget	200	same for both methods
baseline AUROC	0.81	supervised
SSL AUROC	0.83 ± 0.02	gain is uncertain

What the result means

Claim improvement only when SSL beats the matched supervised baseline across repeated label subsets.

Evaluation without label leakage: checking the result

Main idea

The unlabeled pool may contain inputs from validation entities, but their labels and grouped identities must not influence training decisions.

What to check

paired gain with uncertainty across seeds and label subsets

Limit

tuning pseudo-label thresholds on the final labeled test set turns test labels into training information

Evaluation without label leakage: implementation

1. split by deployment entity
2. freeze a labeled test set
3. match the label budget
4. repeat seeds and label subsets
5. report calibration and subgroup results

Check the result

Check paired gain with uncertainty across seeds and label subsets on training and held-out examples.

Common mistake

tuning pseudo-label thresholds on the final labeled test set turns test labels into training information

Evaluation without label leakage: limitations

Example

comparing methods under a fixed annotation budget

Explain the calculation and check paired gain with uncertainty across seeds and label subsets, using a small example.

Common mistake

tuning pseudo-label thresholds on the final labeled test set turns test labels into training information

What to do

Claim improvement only when SSL beats the matched supervised baseline across repeated label subsets.

Evaluation without label leakage: check your understanding

Question

Which labels may be used for tuning, and which must remain untouched?

Answer

The unlabeled pool may contain inputs from validation entities, but their labels and grouped identities must not influence training decisions.

Decision. Claim improvement only when SSL beats the matched supervised baseline across repeated label subsets.

Spend the labeling budget where it changes the boundary

Decision

Start from a supervised learning curve, then label the regions with high uncertainty and high expected influence.

Evidence

Report performance against the number of newly labeled samples, not only against training iterations.

Stop condition

Stop pseudo-labeling when new labels reinforce one class or no longer improve a label-only validation set.

Check the assumptions behind a semi-supervised gain

Decision

Match every semi-supervised method to the cluster, smoothness, low-density, or multi-view assumption it needs.

Evidence

Perturb graph construction, confidence thresholds, and unlabeled-class balance separately.

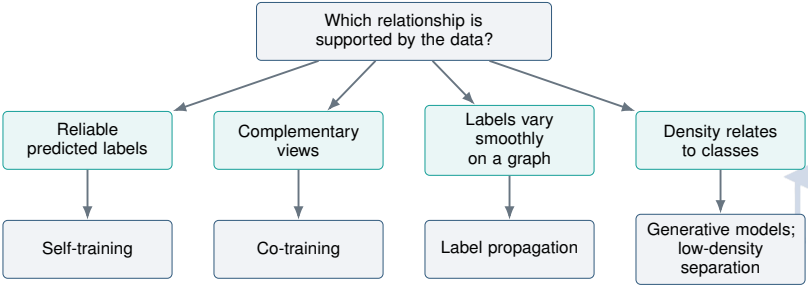
Stop condition

Reject the gain if it disappears under a nearby plausible setting or under a shifted unlabeled pool.

Compare methods by their structural assumption

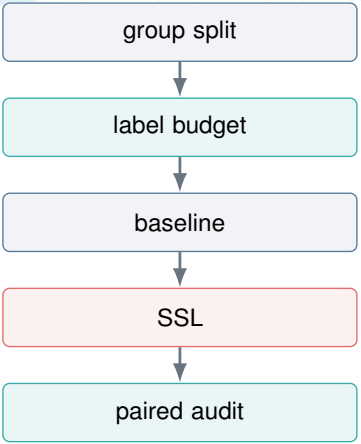
Method	Unlabeled-data signal	Main diagnostic
self-training	confident predictions	pseudo-label precision
co-training	cross-view agreement	error correlation
generative	input density	density and class alignment
S3VM	low-density boundary	margin-region density
graph	local smoothness	edge purity and stability
constrained clustering	pairwise constraints	violations and stability

Choose a semi-supervised method by its assumptions



Compare with a supervised baseline using the same label budget and independent labeled evaluation data.

A leakage-safe semi-supervised experiment



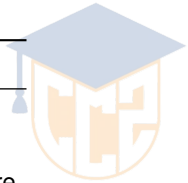
1. split by entity before using any labels
2. freeze a labeled test set
3. fit a matched supervised baseline
4. fit SSL without test-label access
5. repeat label subsets and random seeds
6. report paired gains and calibration

Unlabeled inputs may enter training; final labels may not.

Measure benefit relative to annotation cost

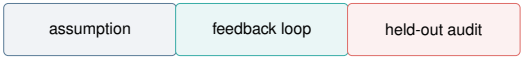
$$\text{value}(b) = \mathbb{E}[\text{score}_{\text{SSL}}(b) - \text{score}_{\text{sup}}(b)]$$

Finding	Interpretation	Action
gain across budgets	assumption is useful	consider deployment
gain only at one seed	unstable	repeat and diagnose
negative gain	assumption mismatch	use baseline or label more



Stress-test pseudo-label feedback

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- audit precision by class and round
- repeat threshold and graph sweeps
- track how many labels change
- compare in-domain and shifted pools
- include a no-unlabeled-data ablation
- stop on independent validation evidence



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Six takeaways to retain

1. unlabeled data provide structure, not class meaning
2. keep a matched supervised baseline
3. calibrate before pseudo-label selection
4. treat graph construction as model design
5. report harm as well as gain
6. prefer new labels when the assumption is weak

Exit ticket: approve or reject unlabeled data

Prompt

Choose one method. State its structural assumption, one tuning parameter, one independent diagnostic, and one condition that would make you reject the unlabeled pool.

Case

A factory has 300 labeled failures, 50,000 unlabeled sensor windows, a changing operating regime, and a fixed budget for 100 additional expert labels.